

2014 ONSC 87 (CanLII)

**SUPERIOR COURT OF JUSTICE**

**HEARD:** June 17, 18, 19 20, 24 and 26, 2013

[1] The plaintiff, Helen Girard, was 72 years of age on November 26, 2007 when she required medical treatment at Windsor Regional Hospital following a slip and fall as a result of which she sustained two Colles' fractures to her wrists, the right worse than the left. This was confirmed by x-rays taken following her admission to hospital.

[2] Both wrists were casted with plaster of Paris splints, applied by Doctors Dawodu and Steyn.

- [3] Because the two wrists were reduced internally, she was administered Propofol, a drug which induces a form of conscious sedation such that while she was aware of the manual reduction of each wrist, she did not experience any pain.
- [4] Upon completion, she was discharged home in the normal course, with an appointment to re-attend at the Fracture Clinic the following day in the early afternoon.
- [5] However while still in the hospital she complained of pain, especially in the area of her right wrist and received a pain medication. In the opinion and experience of the attending nurses and doctors, pain after a cast has been applied, is an expected outcome.
- [6] The following day the splints were removed and appropriate full casts were applied. At that time she complained that her right arm was burning intensely which was interpreted by the treating staff to be the result of an inflammatory response to the affected area which can tighten the cast and cause tissue damage and, in extreme cases, necrosis.
- [7] However it was ultimately determined that she had sustained full thickness (formally known as a third degree) tissue damage as a result of an exothermic burn to her right forearm.
- [8] At the outset of the trial, counsel advised me that they had agreed that the plaintiffs' damages would be the sum of \$50,000 if successful, plus costs.

THE EVIDENCE:

**i. Helen Girard**

- [9] Ms. Girard was leaving her daughter's home next door to where she lived when she slipped and fell and sustained the fractures in question.
- [10] After falling, she went home, and arranged to have an ambulance take her to Windsor Regional Hospital.
- [11] Upon arriving she was taken straight to the x-ray department and then to a resuscitation area where she was examined by the defendants Dr. Steyn and Dr. Dawodu.
- [12] The internal reduction procedure was explained to her by Dr. Steyn who attended to the left wrist while Dr. Dawodu treated the right one.
- [13] She recalled being administered the medication, Propofol, which masked the pain she would otherwise have felt during the course of the manual reduction of her two fractures.
- [14] When the Propofol wore off and she came to her senses, she felt a burning sensation in the right hand and arm, like it was on fire. She requested her daughter Valerie Gagnon summon a doctor. In response Dr. Dawodu returned and informed her that this kind of sensation was to be expected from the formation of a cast.

- [15] She denied that she was told anything by him about what she should do for it and denied there was any discussion about the risks associated with the procedure itself.
- [16] Before being discharged, she was provided with an appointment card to attend the Fracture Clinic the following day, November 27, 2007 but she denied receiving a pamphlet outlining the warning signs that might arise post-operatively or any additional pain medication.
- [17] The next morning she still experienced the intense burning sensation of the night before although it had diminished somewhat.
- [18] At the Fracture Clinic presided over by Dr. Fleming, an orthopaedic specialist, he told her that he would apply full casts on each wrist after removing the temporary splint casts. At that time he observed five blisters between the area of her elbow and her thumb, on the right side which she described as being white and watery, approximately one inch in diameter.
- [19] Dr. Fleming described these as routine fracture blisters. Because they had not burst they were sterile. He applied the cast over them.
- [20] About three days following her discharge she noticed that her right arm was leaking a smelly fluid from a blister under the cast. When she advised Dr. Fleming on December 10 he cut out a “window”, from a piece of the cast midway along it in order to better view this. He confirmed that it was a pustule leaking a green fluid. He prescribed antibiotics and referred her to the Community Care Access Centre (“CCAC”) for further treatment.
- [21] By the time of her appointment on December 17, CCAC was visiting her on a daily basis to change dressings and apply antibiotics. When Dr. Fleming removed the cast he could see that her right arm was red and black. He prescribed further antibiotics and consulted with a wound specialist who examined her arm and then discharged her home again. The CCAC continued its daily visits to dress and attend her wound until March 2008.
- [22] On December 24, 2007, an appointment was arranged for her to see Dr. Sion, a plastic surgeon who indicated that she might require a skin graft after the fracture healed. By the time her treatment with him ceased in March 2008, there had been substantial improvement to the affected area. Her only concern was a tightness of the skin surface which felt leathery; it was neither resilient nor flexible.
- [23] Prior to this incident she was completely mobile and self-supporting but was rendered severely restricted from many of her prior activities because of the casts on her wrists. However, she admitted that the only real change in lifestyle since the accident is the fact that her daughter now cuts the grass for her.
- [24] She admitted that if she had any significant concerns or problems arising from the fracture or the cast application, she could have called her family doctor. However she never did.

**ii. Valerie Gagnon**

- [25] Ms. Gagnon accompanied her mother to Windsor Regional and while in the Emergency Room, Dr. Dawodu arrived and referred her for an x-ray. Dr. Steyn's examination confirmed the diagnosis of bilateral Colles' fractures to her mother's wrists which would require casting for which she would be administered an injection to diminish the pain associated with the procedure.
- [26] After the casting procedure was completed her mother complained of pain and a burning sensation in her right arm which was hot to the touch. In response to this Dr. Dawodu said that the tensor bandage which had been wrapped around the plaster splint, was probably too tight so he loosened it and re-banded the splint.
- [27] The following day at Dr. Fleming's Fracture Clinic, she observed large blisters between the elbow and the wrist when the splint cast was removed from the right wrist and arm.
- [28] Dr. Fleming explained that these were fracture blisters which, because they were unbroken, and sterile, enabled him to apply a full circumferential cast over them.
- [29] Between November 27 and the next visit of December 10, her mother's arm became smelly and green pus was leaking from under the cast. When Dr. Fleming cut a portion of it away, near the elbow he could see that the skin underneath it was red and blackish with a green-yellow discharge for which he prescribed medication. After that date, her mother's right arm was slowly healing.
- [30] On December 17, because of the continuing complaints from her mother along with the presence of leakage under the cast, he elected to remove it and treat the infection, for which he called in a wound specialist. Once the cast was removed, the wounds were cleaned and dressed.
- [31] At her December 24 appointment with Dr. Sion, the plastic surgeon, he observed that it appeared her mother had sustained a full thickness (third degree) burn.

**iii. Dr. Christopher Scilley**

- [32] He is a certified plastic surgeon practising with the London Health Sciences Hospital ("LHSH") specializing in trauma and reconstructive surgery since 1989. I accepted his qualifications as an expert to give opinion evidence as to the complications arising from a cast application and whether a burn is caused by a cast application.
- [33] In his view pressure problems from casts are far more common than burn issues. In his experience he has never had a referral relating to an exothermic burn nor had he ever seen one, prior to being consulted about this case.

- [34] The distinction between a burn and a pressure injury comes down to tissue injury which will occur when there is a lack of circulation in the tissue. Tissue response is the same whether there is a burn or cast application pressure, in that it leads to tissue death. However, one is unable to determine which one of these two would be the effective cause.
- [35] Because the right wrist was more seriously injured it required a more diligent and forceful internal reduction which could cause more swelling than on the left side.
- [36] Blisters develop as a result of the inflammatory response which involves the migration of healing and treating cells from surrounding tissues and capillaries.
- [37] Two types of cells migrate to a wound; those which heal and those which fight infections. It is the migration of these cells to the wound area which causes the swelling. The more serious the wound, the greater amount of swelling because of the presence of larger numbers of healing and infection-fighting cells which have migrated to the wound site. In his experience a Colles' fracture is frequently associated with swelling and the appearance of blisters.
- [38] From the records he noted that blisters were found near her elbow, as opposed to the fracture site, which he found to be unusual as was the string of blisters running from the elbow to the wrist.
- [39] In his experience a patient's complaint of burning from a cast involves a highly subjective concept of pain which is very difficult to quantify since there is no scale to measure it by.
- [40] Once the source of a thermal burn is removed the pain will decrease but it will not necessarily disappear. In the days following, the wound pain will improve due to the inflammatory response and as the swelling subsides this reduces the pressure from the cast and the resulting pain.
- [41] Cast pain can be caused either by the heat created from mixing the Plaster of Paris and water or because the cast is too tight.
- [42] With respect to the former, the warmer the water temperature used to mix the cast, the more heat will be generated. A cast temperature of 100°C (212°F) will cause an immediate burn whereas 60°C (140°F) will be less severe.
- [43] A cast will normally heat up within five or ten minutes of application and will be cool after about 20 minutes. Therefore a burn would occur in the first 20 minutes.
- [44] If the source of the pain is cast pressure this can be alleviated by splitting it open. The degree of cast pressure and the amount of resulting damage which might be caused, will very much depend on the amount of pressure applied in forming the cast, initially.

- [45] Pressure like heat is a highly subjective concept such that there is no way to measure the amount of pressure experienced in the formation and application of a cast. However if the pain experienced by the patient is out of proportion to the nature of the injury, this should raise an alarm.
- [46] After a cast is applied, a patient will often complain of worse pain than before it was applied but this will decrease in the ensuing couple of hours. If there is no relief and the pain increases, the patient should then return to the hospital.
- [47] Whether the skin damage is caused by a burn or cast pressure, will become immediately apparent. If it is a burn, tissue damage will occur within seconds, or minutes, depending upon the temperature of the cast whereas if by cast pressure, then loosening the tensor bandage which is wrapped around the temporary cast splint, should decrease the pain. In either event, however, there is no need to keep the patient in the hospital and a timely discharge is always appropriate.
- [48] At the LSH Fracture Clinic which operates three days a week, he sees approximately 100 people a day. Over his 30 years' experience he estimated that he has been involved in as many as 30,000 different cast applications. During all that time he recalled that he might have seen only one or two thermal burns which are not a common phenomenon. Similarly he has never heard any of the other doctors on his team refer to a thermal burn.
- [49] Any infection arising from either burns or pressure damage is properly treated with a topical antibiotic.
- [50] Even with the presence of unbroken blisters which are sterile, he will apply a cast because they indicate only a superficial injury. His practice is to pop them and then apply a dressing but in his experience, while approximately 50% of burn surgeons do likewise, the other 50% do not.
- [51] The medications administered to the plaintiff here are fast-acting drugs which wear off quickly which permits an early discharge home with prescribed pain pills.
- [52] In his opinion Dr. Fleming who was the treating doctor would be in a better position to assess the plaintiff because he attended with her at the fracture clinic the next day and was on the case for several weeks. This provided him with a much better opportunity to assess her on an ongoing basis.
- [53] While a full thickness injury is the modern restatement of the older phrase "third degree burn" which refers to all three layers of the skin being damaged, this may not be apparent in the first 24 hours, as the wound is developing.
- [54] In his opinion the plaintiff's injury was caused by cast pressure and not a burn and he reiterated that over his many years of experience he has seen numerous cast pressure injuries but never a cast burn. But because it is possible that the injury may be attributed to both, one cannot say definitively whether it is a pure burn or a pure pressure injury.

- [55] When she saw Dr. Fleming in the Fracture Clinic the following day the purpose of the casting was to stabilize the injury; that is, to fixate it. Because one has to be sure that the cast is tight enough to maintain the stability of the fracture but not so tight as to injure it further, the cast application is a judgment call of the doctor, based upon his wisdom and experience.
- [56] It is especially important to consider the fracture stability issue before deciding to remove a cast prematurely because to do so compromises that stability. The splint that was initially applied was not circumferential; that is to say it did not cover right around the arm as the cast would. Because of the nature and severity of the right wrist fracture, it was very important to ensure that it was fully stabilized which is also a medical judgment call balancing the need to maintain stability against the complaint by the patient of pain.
- [57] Unlike hindsight which is always clear cut, an Emergency Room doctor does not enjoy this luxury in the assessment and treatment of a patient who presents herself there.
- [58] In his opinion where a patient complains of pain 45 minutes after being casted, the burn damage will already have occurred and removing the cast at that point would not make any difference in the severity of the burn.
- [59] One might assume that if the two reductions here were done simultaneously using the same water temperature and the same cast application techniques, that the outcome would be the same. However there are a number of variables which affect this including; the difference in the severity of the injury between the left and the right wrist; whether or not the skin of the patient is old and sensitive, elevation of the arm and other post-surgical treatment, all of which could affect the difference in outcomes.
- [60] When he reviewed Dr. Fleming's clinical notes with respect to the issue of infection, the right arm was noted to have significant breakdowns with fracture blistering. Because there was no suggestion of infection in any of the medical notes and records and the photographs cannot confirm that the injury was due to an infection, it is reasonable that he treated her arm with topical antibiotics, as he considered appropriate.
- [61] In his opinion Dr. Fleming's treatment of the plaintiff was appropriate.
- [62] Her delayed response to pain could arise because there was a residue of the two drugs in her system that the plaintiff was prescribed during the course of the reduction procedures which could alter her perception of the severity of her pain, producing a blunted pain reaction. Were it not for the medications, she would have experienced much more pain.
- [63] Again, in his opinion, one cannot say whether this was a thermal issue or a cast pressure issue.

**iv. Dr. Omotayo Dawodu**

- [64] The defendant Dr. Omotayo Dawodu is a certified medical practitioner whose experience includes primary care as well as emergency medicine. He has been certified to practise

medicine in Alberta and Ontario and while serving at Windsor Regional Hospital, he attended the plaintiff on November 26, 2007 when she was admitted with fractures to her wrists.

- [65] X-rays confirmed that she had sustained bilateral Colles' fractures of her wrists. The right, a comminuted fracture in several places, was more severe than the left.
- [66] He stressed the importance of stabilizing the fractures, especially the right wrist because if it was not, it could not be reset. Rather, it would necessitate a trip to the operating room to undergo an open reduction with the insertion of a plate and screws. All of this would carry the potential for infection.
- [67] She was prescribed Fentanyl and Propofol in order to achieve the state of "conscious sedation" where, while she would remain awake, she would be impervious to pain while her fractures were being reduced. He described this as a technique which is used worldwide.
- [68] In the short term it is important to reduce the fractures, because the longer the period of time the fracture remains unreduced, the greater will be the pain experienced by the patient as well as stiffness which can occur later and thereby produce a different outcome.
- [69] Prior to proceeding, he explained the procedure and its risks and benefits to her and prescribed the medications which would be administered by Dr. Steyn. He received her consent to proceed.
- [70] While Dr. Steyn was able to reduce the left fracture, Dr. Dawodu, because of the comminuted nature of the right wrist fracture, required the assistance of Dr. Steyn who held her right arm in place while Dr. Dawodu applied the necessary manipulation in order to achieve a reduction, whereby the various pieces of bone were rejoined.
- [71] At this point a back slap splint was applied by him to the right wrist in order to stabilize the reduction and minimize the amount of swelling which would inevitably occur with the application of the splint cast.
- [72] He is in agreement with Dr. Scilley that the fracture would necessarily cause swelling, due to the ongoing inflammatory process which involves the migration of cells to the injury site, both to heal and to clean the wound. This migration causes the swelling in addition to that arising from the application of the splint cast.
- [73] He discharged the plaintiff home at approximately 22:38 with an appointment to return to the Fracture Clinic the following day and he stated that the appointment card together with a two page document outlining the warning signs to watch for, would have been provided to the plaintiff in an envelope by one of the nursing staff.
- [74] Both before and after the reductions, until her discharge, the plaintiff was hooked up to a number of monitors to check her vital signs as the evening progressed and at all times,



they indicated that her condition stabilized afterward and that there was no indication of stress throughout.

- [75] Consistent with the standard of care, Dr. Steyn was assisted by a Nurse Practitioner, a recording nurse and a respiratory therapist, all of whom are part of the standard of care practiced for the conscious sedation procedure which renders the patient immune to the pain which would be generated by the fracture reductions, although she would be aware of the procedure occurring.
- [76] Splinting was described as a standard practice whereby Plaster of Paris is mixed with water to form the splint which would cover approximately 60% of her right wrist and arm, in order to ensure the integrity of the reduction. However caution must be exercised to ensure that the temperature of the water is not too warm because, the warmer the water, the more exothermic heat is generated when it is mixed with the gypsum material to form the Plaster of Paris. For this reason tap water is used which is tested by dipping one's fingers into the water bowl before it is applied to the gypsum material in order to ensure that its temperature is not too warm.
- [77] Once mixed, the Plaster of Paris is then placed over the splint at which time an exothermic reaction will commence during the hardening process. A tensor bandage is then wrapped around the splint before hardening, in order to keep it in place. And lastly, a cotton stocking is applied over the hand and up to the elbow in order to absorb the pressure and to keep the splint material stable as well as to dissipate heat from the Plaster of Paris.
- [78] While he has applied thousands of casts over many years, he has never encountered a thermal burn.
- [79] Dr. Dawodu stayed with the plaintiff both during and after the procedure. When she opened her eyes and was able to communicate spontaneously, this demonstrated that the splint was in place.
- [80] At this point, if she had been experiencing discomfort, her vital signs as displayed on the various monitors, would change. They did not and in fact on three successive occasions during the course of the time she remained in the Emergency Department, up to her discharge at 22:35 hours the blood pressure, heart rate and pulse did not display any increases which would be consistent with abnormal pain or discomfort.
- [81] Just before discharge, this defendant conducted a neurovascular assessment and arranged for post-procedure x-rays which confirmed that a satisfactory union had been achieved. At that point he explained to his patient, the fracture and casting procedure that would be taking place the following day at the Fracture Clinic and alerted her to the warning signs contained in the document entitled What To Do If Problems Arise. Should she experience extreme pain, he recommended extra strength Tylenol.
- [82] She was also provided with some general instructions about not driving or riding a bicycle, and to have responsible family members nearby to assist her as required.

- [83] Once the final assessment had been completed, together with a last check on her vital signs, she was then discharged.

**v. Carrie Pillon**

- [84] Ms. Pillon, R.N., and now a Nurse Practitioner, has worked in the Emergency Room at Windsor Regional since 2007 and has attended many fractures over her career but has never experienced a complaint of a thermal burn from a cast.
- [85] Given the passage of time, she had no direct recollection of the events and relied on the charts which confirmed that on arrival, the plaintiff was diagnosed as having sustained bilateral wrist fractures, with the right worse than the left and subjectively, the patient stated that this was the worst pain she had ever experienced.
- [86] This diagnosis was confirmed by x-rays which were taken at the instruction of the Emergency Room physician who assessed her before the casting procedure commenced.
- [87] She confirmed that the box on the admission document indicating whether verbally informed consent was obtained, was not completed but in her opinion this would have been discussed with the patient otherwise, the procedure would never have proceeded.
- [88] Furthermore, consistent with nursing practice, where one only charts to exceptions, unless the pain reported by the patient was something out of the ordinary in the experience of the nursing staff, it would not be charted.
- [89] She also confirmed that the records indicated that at 22:30 the plaintiff was reassessed by Dr. Dawodu and then discharged home with her daughter, with an appointment for the following day at the Fracture Clinic. The last check of her vitals which all displayed normal, was made, before she was unhooked from the monitors at approximately 22:35.
- [90] She stated with certainty that she or other members of the nursing staff would have provided the post-surgical documentation either to the plaintiff or a member of her family together with the appointment card for the Fracture Clinic the next day, which the plaintiff confirmed receiving.
- [91] Furthermore, if in the opinion of the nursing staff the plaintiff should not have been discharged, the doctor's order would be countermand. In the career of the witness, she has done this approximately 80 times.
- [92] She also confirmed that following the reduction the swelling would increase because of the inflammatory process taking place together with the additional tension in the fracture splint arising from the necessary pulling and twisting that had to be undertaken in order to initially reduce the fracture.

**vi. Dr. Dana Fleming**

- [93] Dr. Fleming has been a certified specialist in orthopaedic medicine for 30 years during which time he has applied thousands of casts. In addition to his practice he is presently the Musculoskeletal Coordinator at the Schulich Medical and Dental School at its Windsor Campus, where he teaches and tests students on, among other things, cast complications.
- [94] He was on duty at the Fracture Clinic on November 27, 2007 when he first came into contact with the plaintiff.
- [95] A review of her x-rays confirmed that she had sustained the bilateral wrist fractures, which he described as a rare occurrence. Both had been reduced and splinted the day before. The right was more severe, being an angulated 45 degrees at the dorsal which is the bone at the top of the wrist.
- [96] In his opinion the treatment afforded to her the previous day was appropriate.
- [97] The splints on her wrists were removed and his examination revealed multiple small fracture blisters resembling bubbles, in the upper right arm which resulted from swelling in the arm caused by the inflammatory process. He described these as superficial in nature which affect only the top layer of skin. In his experience a cast can be applied directly over them which is the normal and accepted practice.
- [98] He was not concerned about their presence and estimated that approximately 20% of people who sustain fractures, experience this phenomenon. Because they had not broken, the skin underneath was sterile.
- [99] While the plaintiff was complaining of pain at the fracture site on the right side, he saw no reason not to proceed to stabilize it. And further, there was nothing further surgically required at this point because the fractures had been appropriately reduced the day before as confirmed by x-ray.
- [100] She was therefore re-casted by the cast technician.
- [101] In his opinion burning is a common occurrence arising from fracture casting which is caused by the sensory nerves reacting to the cast.
- [102] The preferred approach in his opinion is to cast fractures, otherwise surgery is indicated which carries a 20% chance of infection. Furthermore to remove a splint before union has been achieved would, with medical certainty, require the patient to be taken to the Operating Room to undergo an open reduction with the internal fixation of a plate and screws or alternatively, an external fixation which requires screws being inserted into the affected area through the wrist. Infection becomes in his opinion a real possibility.
- [103] He described a cast burn as the chemical reaction caused by mixing gypsum material with water to form the cast. However, in his experience, this is a rare occurrence and he last

saw one in 1983. Therefore he did not consider a cast burn as an issue when he saw the blisters.

- [104] The three factors in a Plaster of Paris splint which are potential sources of heat and burns are the temperature of the water which should not exceed 24°C, the cast thickness should be no more than eight plies of wrapping (although the practice at his hospital is to utilize five) in order to minimize heat from the chemical reaction radiating inward to the skin under the cast and inadequate ventilation of a newly applied splint or cast.
- [105] Because of the acceleration of heat caused by using warm water, it is his practice always to use cold water which while a little slower, is safer. The difference between the two approaches is only approximately two to three additional minutes where the attending physician might be required to stand and keep the splint in place while applying pressure, as it hardens.
- [106] Nevertheless even if he had concluded that she had sustained a burn, he would not have treated her injury in any different way.
- [107] In the plaintiff's next visit of December 10, he observed a significant breakdown of the skin together with more skin burn than he would have expected and accordingly he "windowed" the cast by cutting a hole through it in order to see the skin of the arm, underneath.
- [108] He did not feel it was appropriate to remove the cast which would compromise the reduction since at this point in time, only two weeks had passed.
- [109] He noted a purulent smell of the cast which in his view indicated that the dressing might be infected, but not her arm. He prescribed an antibiotic and instructed the CCAC to attend daily to treat the arm with dry dressings. The wound was not, in his opinion infected.
- [110] While he was unable to state with any certainty that there was infection, he nevertheless prescribed the low dose antibiotic, Amoxicillin, 250mg four times daily.
- [111] At the follow-up visit on December 17, approximately four weeks post- reduction, he removed the cast on the right arm and observed a patch of skin loss which extended from the thumb up to the elbow, consistent with the burn and somewhat square in shape. It resembled the outline of the plaster splint. In his experience this was very rare.
- [112] At that point he changed his diagnosis from a fracture blister to a burn.
- [113] He did not consider this to be caused by pressure from the splint because the muscle underneath was not been damaged and in 30 years of practice he had never seen a pressure injury resembling this one.

- [114] When the cast or splint is first applied with padding on top, the number of wraps around the arm can be significant because too many will cause the heat which is generated from the chemical reaction of the plaster with water, to radiate inward to the arm as opposed to being dissipated through the top dressing. Therefore multiple layers of bandage wrapping on top of the splint are not recommended. However there was no evidence here to suggest that there were too many layers of wrap applied.
- [115] On the plaintiff's final visit of January 9, 2008, he noted the fracture was now well healed but the skin issues continued which he described as a full thickness burn to the mid portion of the right arm. She had been earlier referred to Dr. Sion, a plastic surgeon, for debridement and a possible skin graft.
- [116] His retrospective diagnosis is a plaster burn with scarring.
- [117] In his experience skin damage can occur in as little as 20 minutes, from the initial application of undue pressure but he was unable to estimate the length of time that would be required to cause a full thickness burn. The consequences of undue pressure arising from a cast fixation is more common than a thermal burn but nevertheless an injury from a thermal source will have a pressure component to it.
- [118] Tissue damage can only be determined by removing the splint but after 15 hours, a thermal burn and pressure damage, are indistinguishable.
- [119] While he acknowledged that despite appropriate care being administered, a poor result had been achieved. It was not obvious as to what happened but he suspected this to be a plaster burn which had occurred prior to him seeing the plaintiff.

**vii. Dr. Chris Steyn**

- [120] Dr. Steyn, duly certified to practise medicine in Ontario assisted Dr. Dawodu by casting the plaintiff's left wrist on November 26, when she attended Windsor Regional Hospital.
- [121] He too stated that, having applied several thousand splints and casts in his practice experience, patients will complain of a burning or fire-like sensation after a cast application, which was not unusual.
- [122] He administered the Propofol to the plaintiff, to induce the conscious sedation state so that her fractures could be reduced without her being sensitive to the pain associated with the procedures.

**viii. Dr. Robert Dunlop**

- [123] Dr. Dunlop was proffered by the defence as a medical expert on the standard of care concerning Colles' fracture treatment with particular reference to the application of splints and casts and managing any complications; secondly, to express an opinion of the

cause of the plaintiff's wound and whether any other treatment would have made a difference.

- [124] After reviewing his qualifications and hearing submissions from counsel, I accepted him as a medical expert as outlined.
- [125] Following his review of the medical records, he concluded that the cause of the skin injury to the right arm was a thermal burn from the application of the splint plates. However, of importance, an earlier diagnosis of a full thickness burn would not have prevented the injury in this case because the damage occurred within the first five minutes.
- [126] In his review of the record and the treatment provided by Dr. Dawodu, Dr. Dunlop found nothing that fell below the standard of care and further, he met the standard of care in discharging the plaintiff.
- [127] Similarly, in his opinion Dr. Fleming met the standard of care with respect to his treatment of the plaintiff from November 27 onward.
- [128] He agreed that when Plaster of Paris is mixed with water it causes heat. Using clean and cooler water will diminish the amount of heat produced but will lengthen the hardening time of a splint or cast. Therefore casting becomes a delicate balance using the doctor's skill and experience between generating too much and too little heat; this will affect the rate at which the casting material hardens.
- [129] Furthermore the majority of fracture procedures in hospitals face the same problem where a patient complains within ten minutes of being casted. Doctors apply a value judgment on what to do and the patient is assessed on the basis of that doctor's experience as well as input from colleagues. In the end, however, it all comes down to a value judgment by the individual doctor.
- [130] The most common cause of a burn arises from the fact that because the patient is in a state of conscious sedation during the reduction and cast application and is unable to subjectively determine the degree of heat generated whereas a conscious patient would immediately feel heat and if excessive, complain about it.
- [131] Even with the best of care an exothermic burn cannot be avoided because of the inevitable production of heat when water is mixed with the gypsum material employed in the cast.
- [132] By the time a cast is hardened, the maximum heat has peaked and it begins to cool. All of this occurs within just a few minutes because of the underlying object to have the plaster harden as quickly as reasonably possible, in order to avoid losing the reduction.
- [133] In his experience while patients will complain of heat from a plaster reduction, it is difficult to tell whether this is caused by the reduction itself or the exothermic reaction.

- [134] Because the heat is generated immediately and then diminishes rather quickly, removing the cast 15 or 20 minutes after its application would make no difference if the patient complains of heat, because by that time the burn will already have occurred.
- [135] In the case at hand the splint was applied at approximately 21:45 while the plaintiff was under conscious sedation, and 40 minutes later she complained of the burning. In his opinion this meant that the burn had already occurred and that removing the cast before seeing Dr. Fleming would make no difference.
- [136] He agreed with the previous medical opinion that the presence of fracture blisters in Dr. Fleming's Fracture Clinic the following day, was all part of the inflammatory healing process. They were of no medical concern and casting over them is normal.
- [137] While the application of either a splint or a cast requires that pressure be applied to the reduction site, Dr. Dunlop has never seen excessive pressure cause a full blown thermal burn.
- [138] He teaches medical students that initially a splint should be used in preference to a cast because it generates less pressure. Furthermore if necessary, the tensor wrap around the splint can be removed or reduced to allow heat to dissipate through the top of the arm, as opposed to being directed internally to the patient's arm.
- [139] When comparing the Windsor experience with that of Hamilton where he practises, Dr. Dunlop stated that the methodology employed in splinting and casting here was superb and he expressed amazement that the plaintiff was accommodated at the fracture clinic within 24 hours whereas in Hamilton she might have had to wait for one or two weeks.
- [140] He agreed that the comminuted fracture of the right wrist was more severe than the left, because of the many pieces of bone that had to be pieced together which makes reducing more difficult.
- [141] Nevertheless the treatment involving the reduction as well as the splint and casting procedure was appropriate and met the standard of care. His opinion was bolstered by the fact that the post-reduction x-rays demonstrated that a good union had been achieved which is consistent with meeting the standard of care.
- [142] The splinting materials used by Dr. Dawodu were appropriate as was the technique he utilized and his post-reduction care and the discharge of the plaintiff all met the standard of care.
- [143] Noting that Dr. Fleming had not yet suspected an exothermic burn, Dr. Dunlop opined that even if a burn was suspected, one would still apply a cast because the first order of business is always to stabilize and protect the fracture, especially a comminuted one.

- [144] Dr. Fleming's removal of the cast on December 17 to treat the wound was reasonable as was his consulting with a wound nurse who prescribed 250 mg of Cloxacillin, four times daily.
- [145] Pain is an inevitable outcome of a fracture and in his opinion approximately 99% of patients who arrive at hospital for treatment of their fracture, are sent home in pain.
- [146] Further there is a marked significance to be placed on the fact that her pre-sedation blood pressure, pulse and respiratory rates were the same as they were post-reduction which indicates in his experience that it is most likely that she was not experiencing too much pain because pain itself will cause these vital signs to increase.
- [147] Dr. Dunlop has performed more than 5,000 Colles' fracture cast applications and repairs but he has never encountered a full thickness burn caused because of undue pressure.
- [148] In his opinion a bad outcome resulted from the procedure employed but not because of poor or substandard care rendered to the plaintiff. Good and appropriate care can, in his experience, sometimes result in a bad outcome.
- [149] He did not consider it to be significant that Dr. Dawodu could not recall, after all the years that have intervened, giving medication, because of the existing hospital medication protocols that would have to be followed when prescribing Fentanyl, which is a narcotic. Control of the medicine cabinet where it is kept, in a hospital setting is maintained by the nursing staff and not the doctors. If Dr. Dawodu prescribed Fentanyl he would be required to sign off on the requisite prescription sheet, otherwise none would have been issued.
- [150] To remove a cast to treat what was thought to be a possible (but not certain) infection, would be, in his opinion, a huge judgment call because medically speaking, the bottom line always remains that one does not want to lose the reduction. Removing the cast could compromise it and make a visit to the Operating Room for an open reduction with a consequent risk of infection, inevitable.

#### FINDINGS:

- [151] The two issues to be considered are firstly, did the treatment by Drs. Dawodu and Fleming meet the standard of care and if not, did the care of either or both cause or contribute to the plaintiff's injury.
- [152] From the evidence and the record before me I find that both Drs. Dawodu and Fleming met the standard of care.
- [153] Dr. Dawodu's involvement with the plaintiff took place in the Emergency Room when he attended to the diagnosis and treatment of the fracture to her right wrist. It is common



ground that the treatment of the left wrist which was a lesser injury than the right, was appropriate.

- [154] Dr. Dawodu firstly arranged for an x-ray of both wrists which confirmed the two Colles' fractures with the right, being a comminuted one, was more severe.
- [155] With his many years of skill and experience Dr. Dawodu applied a splint to the right wrist in what could be described as in textbook fashion which, as confirmed by the post-procedure x-rays, was reduced and casted appropriately.
- [156] I accept his evidence that prior to commencing treatment he received the plaintiff's verbal consent to proceed. The Admission Records indicate that the box on the sheet referring to consent, had not been checked off but I can conclude that this was probably an administrative oversight and further, it would be highly unlikely that Dr. Dawodu would have proceeded with treatment, if she had not consented.
- [157] The procedure employed was consistent with that described by the defendant's expert Dr. Dunlop who was very impressed by the fact that the plaintiff would be seen the following day in the Fracture Clinic supervised by Dr. Fleming.
- [158] Prior to discharge Dr. Dawodu assessed the plaintiff and noted that her vital signs which had been monitored at various times during her stay, remained normal, indicating an absence of excessive pain. Monitoring of the vital signs provided perhaps the only objective measurement of pain which is otherwise a highly subjective experience.
- [159] Heat and pain are the expected results of any casting procedure and the plaintiff was advised of this. She and her family were also advised that she should rest at home with her arms elevated on pillows which, her daughter testified they did. This would refute the suggestion that the plaintiff did not receive any advice on post-operative care.
- [160] The plaintiff conceded that her memory was somewhat frail which in the circumstances is understandable bearing in mind that she was focussed on her injury, as well as the pain associated with it and the casting procedure. I also note that she did not remember the post-casting x-rays being taken. All of this would call into question the accuracy of her recollections, when contrasted with those of Dr. Dawodu and Nurse Pillon.
- [161] I accept the evidence of Dr. Dawodu which is supported by Drs. Dunlop and Fleming that a discharge home at this point was reasonable and appropriate. Had the nursing staff felt that the plaintiff should not, in their wisdom and experience be discharged, they would have countermanded Dr. Dawodu's instructions. Nurse Pillon, who has many years of experience, has over the years countermanded a doctor's discharge order approximately 80 times.
- [162] On discharge the plaintiff was provided with a card confirming her appointment the next day at the Fracture Clinic with Dr. Fleming, together with a two page document of post-

procedure instructions. From the evidence there is some discrepancy as to whether she received the first of those two pages and, from whom. However observing that it was approximately six years ago that the plaintiff was treated and that both Dr. Dawodu and Nurse Pillon have undoubtedly treated many hundreds if not thousands of subsequent patients, it is not surprising that neither of them could recall precisely who handed what document to whom. However I am satisfied that it was carried out in accordance with standard hospital operating procedures. I find that it would be unlikely that the plaintiff received the appointment card and the second page of the post-procedures instructions, without being given the first page. To suggest otherwise lacks an air of reality.

- [163] Furthermore as noted earlier, the plaintiff stated that her memory was not all clear.
- [164] A witness may testify from habit as a substitute for actual recollection where he/she can refer to something which is done regularly in their personal or professional lives (see: *Belknap v. Meakes*, [1989] 64 D.L.R. (4th) 452, at pp. 13-14 and *Mirembe v. Tarshis*, [2003] O.J. No. 4753, at para. 1 (C.A.)). Accordingly, I therefore accept the evidence of Dr. Dawodu and Nurse Pillon, accordingly.
- [165] What emerged from the testimony of all the medical witnesses is the importance of saving a reduced fracture. It is clear that removing a cast prematurely and especially immediately following a reduction involving a complex fracture such as the comminuted fracture here, runs a very high risk of displacing the reduction. In that event the only medical recourse is to take the patient to the Operating Room and perform an open reduction with the insertion of a plate and screws or alternatively, apply an external fixator which is set in place with screws drilled into the affected area which would undoubtedly run a 20% chance of infection.
- [166] This no doubt influenced Dr. Dawodu's decision to leave the splint in place although he loosened the tensor wrap somewhat to help alleviate the plaintiff's pain. But as noted earlier all of the medical evidence confirmed that heat and pain are the inevitable consequence of a casting procedure which provides the context for the comment by Dr. Dawodu to the plaintiff that the pain she was experiencing was to be expected.
- [167] The medical evidence also establishes that an exothermic or thermal burn is very rare whereas a pressure burn from a cast or a splint is not uncommon. However several hours afterward, it is not possible to determine one from the other.
- [168] Furthermore I accept from the evidence that a burn will occur within a very short period of time, a matter of a few minutes. Accordingly if there was a reason to remove the cast or a splint (and I am satisfied that there was not, here) then the burn would have already occurred.
- [169] The first person who viewed the plaintiff's arm post-splinting was Dr. Fleming, who concluded that it had been applied appropriately. At that time he noticed the presence of sealed blisters which he diagnosed as fracture blisters, a common occurrence resulting

from the fluid generated by the healing and infection fighting cells of the body migrating to the injury site which causes the blisters. They are sterile if unbroken.

- [170] I accept that casting over them is an accepted practice. Approximately 50% of surgeons will pop them before casting whereas the other 50% do not.
- [171] Either way casting over them as Dr. Fleming did is an accepted practice and comports with the standard of care.
- [172] I find that as the treating physician, Dr. Fleming's care of the plaintiff from her first visit of November 27 up to when she was referred to Dr. Sion, a plastic surgeon, likewise met the standard of care.
- [173] All of the medical opinion confirmed that as the treating physician, Dr. Fleming was in the best position to know what to do. His treatment regime which included windowing the cast on December 10 and prescribing antibiotics and a referral to the CCAC and a wound nurse, were appropriate.
- [174] The direct evidence of a treating physician should be preferred over the opinion of an expert and even the plaintiff's expert Dr. Scilley confirmed that Dr. Fleming was in a better position to diagnose the cause here (see: *Ortolan v. Hotel Dieu Grace Hospital*, [2011] ONCA 456, at paras. 11-13).
- [175] I reject the suggestion he should have removed her cast earlier than he did, for the same reason, namely, to preserve and protect the fracture and avoid a possible open surgical procedure with its attendant risks.
- [176] The burn wound on the plaintiff's right arm was an injury in progress to which Dr. Fleming responded appropriately and provided the basis for the change in his initial diagnosis of fracture blisters to that of a cast pressure or thermal burn, on December 17.
- [177] Nor was Dr. Scilley critical of Dr. Fleming's initial diagnosis of fracture blisters on November 27 and that diagnosis of the wound at that early stage was not possible. In Dr. Dunlop's opinion it would only become obvious over the course of the following weeks when the skin tissue began to slough off.
- [178] I accept the evidence of Drs. Dawodu, Fleming and Dunlop of the correlation between the temperature of the water used to create the plaster cast and the amount of heat generated. The colder the water, the less heat but the longer will be the setting time whereas the warmer the water, the faster the plaster will set but the more heat will be generated. From the evidence, I accept that cold tap water temperature is the accepted and usual method of preparation but I reject the plaintiffs' suggestion that 10°C water must be the standard.
- [179] In advancing this argument, counsel appeared to be suggesting a speculative theory which had no evidentiary foundation and referred to a document which had been marked

as an exhibit for reference purposes only. None of the propositions he advanced, were put to any of the medical experts.

- [180] I am satisfied that the full thickness burn which developed could not have been anticipated by Dr. Dawodu. It was not reasonably foreseeable, and by the time Dr. Fleming saw her, the damage had already occurred.
- [181] Dr. Fleming's diagnosis of the burn was confirmed by Dr. Sion, the plastic surgeon, to whom he referred the plaintiff. Both Drs. Fleming and Dunlop agreed that if the injury was caused by pressure, they would expect to see muscle and nerve damage whereas the plaintiff's wound was confined to the skin and tissues. There was no muscle or nerve necrosis.
- [182] Further, Dr. Fleming and Dr. Sion also confirmed that the shape of the burn which coincided with the splint applied, supports the conclusion that the cause of the burn was thermal.
- [183] In the final analysis, therefore, I conclude that the treatment afforded the plaintiff by the defendant doctors, Dawodu, Steyn and Fleming, met the standard of care. Furthermore, I conclude that the case administered by Drs. Dawodu and Fleming did not cause or contribute to the injury complained of.

#### THE LAW:

- [184] My opinion is bolstered by the principle that physicians are not to be held to a standard of perfection (see *Lurtz v. Duschene*, [2003] O.J. No. 1540, S.C.J.) where Lalonde J. stated that in tortious claims of medical cases, the court does not apply a test of perfection. The physician is expected to exercise a degree of skill and care expected of a normal, prudent physician of comparable training and experience.
- [185] A medical practitioner must exercise a reasonable standard of care and exercise a reasonable degree of skill and knowledge. The standard to be adhered to is not one of perfection but that of the normal prudent practitioner of the same experience and standing. If he is a specialist, a higher degree of skill is required than one who is not so qualified (see: *Crits v. Sylvester*, [1956] O.J. No. 526 (O.C.A.)).
- [186] As was aptly stated by Denning L.J., in *Roe v. Minister of Health et al*, [1954] 2 Q.B. 66, at p. 83, it is easy to be wise after the event and label as negligence something that was only a misadventure. The great benefits created by medical science for society are necessarily accompanied by risks. Every surgical procedure has risks.
- [187] The court must be careful not to rely on the perfect vision which is accorded to hindsight and a doctor's limited ability to peer into the future when determining the course of

treatment or conduct which must be borne in mind to evaluate that doctor's exercise of clinical judgment, fairly.

- [188] A doctor will not be found liable where he/she is found to have appropriately applied his/her clinical judgment and a court should not become involved in questions of assessment having to do with diagnosis or treatment of preference (see: *Lapointe v. Hôpital Le Gardeur*, [1992] 1 S.C.R. 351).
- [189] An error in judgment has long been distinguished from an act of unskillfulness or carelessness. Furthermore a physician is not to be judged by the result.
- [190] In assessing whether a doctor has met the standard of care, the practitioner may testify as to his or her invariable practice and this evidence is admissible as a substitute for present recollection (see: *Belknap*, at pp. 13-14 and *Mirembe*, at para. 1).
- [191] It is not open for the court to choose between two schools of scientific thought, each one of which seems to be reasonable. Where a doctor treats a patient in accordance with a respectable body of medical opinion, even if it is a minority, he will not normally be held liable and negligent. A court's preference for one body of opinion over another is not sufficient to establish negligence (see: *Lapointe*, at paras. 31-33). The standard of care is not established by an expert who testifies as to what he would do in a situation rather than what the standard of care requires. The testimony of that expert does not therefore establish the standard of care nor does it demonstrate that the doctor has breached it.

#### CAUSATION:

- [192] As in any malpractice action the plaintiff is required to prove firstly, that the treatment provided breached the standard of care and that secondly, the defendant physician's care caused or contributed to that injury. That is to say, the plaintiff must show that "but for", the negligent act or omission of each defendant, the injury would not have occurred.
- [193] With respect to the issue of a delayed diagnosis, the plaintiff must prove that any delay caused or contributed to the unfortunate outcome such that if she fails to prove that the unfavourable outcome could have been avoided with a more prompt diagnosis and treatment, then her claim must fail.
- [194] In my view the plaintiff has failed to prove that a breach of the standard of care and secondly, that the care provided to her caused or contributed to her injury.
- [195] Lastly, I wish to comment on the notion of foreseeability, given the allegations of tortious misconduct by the plaintiff in the form of medical malpractice.
- [196] The seminal tort case on the issue of reasonable foreseeability is the decision of the Privy Council in *Overseas Tankship (U.K.) v. The Miller Steamship Co.*, [1967] 1 A.C. 617 (Wagon Mound No. 2), which has been applied recently by the Supreme Court of Canada

in *Mustapha v. Culligan of Canada Ltd.*, 2008 SCC 27, where McLachlin C.J. described the proposition of reasonable foreseeability as follows:

Much has been written on how probable or likely a harm needs to be in order to be considered reasonably foreseeable. The parties raise the question of whether a reasonably foreseeable harm is one whose occurrence is probable or merely possible. In my view, these terms are misleading. Any harm which has actually occurred is “possible”; it is therefore clear that possibility alone does not provide a meaningful standard for the application of reasonable foreseeability. The degree of probability that would satisfy the reasonable foreseeability requirement was described in the *Wagon Mound* (No. 2) as a “real risk”, that is “one which would occur to the mind of a reasonable man in the position of the defendant...and which he would not brush aside as farfetched” (*Overseas Tankship (U.K.) v. The Miller Steamship Co. Pty*, [1967] A.C. 617 (P.C.) at p. 643.

- [197] The court went on in that case to state that “unusual or extreme reactions to events caused by negligence are imaginable but not necessarily foreseeable”.
- [198] Furthermore, as was stated by Hall J. in *University Hospital v. Lepine*, [1966] S.C.R. 561, whether or not an act or omission is negligent must be judged not by its consequences alone, but also by considering whether a reasonable person should have anticipated that what happened might be a natural result of that act or omission.
- [199] As I have concluded, the plaintiffs have failed to establish that there was a breach of the standard of care and having concluded that, it is unnecessary to consider whether any breach caused the damage. Had I decided otherwise, however, I would have concluded that it did not.
- [200] The plaintiffs’ action will therefore be dismissed with costs.

#### COSTS:

- [201] Any discussion on the issue of costs engages s. 131 of the *Courts of Justice Act*, R.S.O. 1990, c. C. 43 (“CJA”) which provides that the costs are in the discretion of the court which may determine by whom and to what extent they shall be paid. As well, Rule 57 of the *Rules of Civil Procedure*, R.R.O. 1990, Reg. 194, lists a number of factors to be considered in addition to the result achieved. However, by rule 57.01(4), none of these factors affect the authority of the court under s. 131 in exercising its discretion in the award of the ultimate costs.
- [202] It is perhaps trite to say that the purpose of costs is to indemnify successful litigants for the cost of their litigation, to encourage settlements, and to discourage and sanction any inappropriate behaviour.

- [203] In general terms there are three levels of costs indemnification, ranging from partial indemnity to substantial indemnity and lastly, full indemnity which represents a complete indemnification at actual rates and which is only rarely applied and in circumstances far more exceptional than those at hand.
- [204] In assessing the appropriate amount of costs to be awarded, I have regard to the following factors in rule 57.01:
- the complexity of the proceeding;
  - the importance of the issues; and
  - the conduct of either party that tended to shorten or lengthen the duration of the proceeding.
- [205] This trial lasted for six days, and while it comprised a medical malpractice claim, it was not in my view, particularly complex.
- [206] Ultimately any assertion for costs must be balanced against the benchmark established by the Ontario Court of Appeal in *Boucher v. Public Accountants Counsel for the Province of Ontario* (2004), 71 O.R. (3d) 291 (Ont. C.A.) where the court clearly stated that the fixing of costs does not begin and end with a calculation of hours times rates, nor was the introduction of the costs grid intended to produce that result.
- [207] This is only one factor of many to be considered in the assessment process, together with those factors referred to in rule 57.01.
- [208] However the ultimate objective is to fix an amount that is fair and reasonable for the unsuccessful party to pay in the particular proceeding rather than an amount fixed by the actual costs incurred by the successful litigant.
- [209] Therefore the overarching principle of reasonableness must remain the primary consideration in the ultimate assessment of the costs and of course, the ultimate discretion afforded to me by s. 131 of the *CJA*.
- [210] I reviewed the Bill of Costs submitted by defence counsel and would, on an assessment basis, settle the issue on the partial indemnity scale.
- [211] In all, the costs proposed approximated \$60,000.
- [212] That said, turning to the principles that ultimately guide me including the principle of reasonableness that, in the exercise of my discretion I determine to be an appropriate amount, I hereby fix the costs payable to the defendants in the sum of \$30,000 for fees and \$5,109.51 including HST for disbursements, for a total \$35,109.51.

Original signed "Justice Gates"  
Richard C. Gates  
Justice

**Released:** January 6, 2014



**CITATION:** Girard and Gagnon v. Windsor Regional Hospital, 2014 ONSC 87  
**COURT FILE NO.:** CV-08-11162CM

**ONTARIO**

**SUPERIOR COURT OF JUSTICE**

**BETWEEN**

Helen Girard and Valerie Gagnon

Plaintiffs

**– and –**

Windsor Regional Hospital, Omotayo Dawodu,  
Omotayo F. Dawodu Professional Corporation, Chris  
Steyn and Dana Fleming

Defendants

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**REASONS FOR JUDGMENT**

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Gates J.

**Released:** January 6, 2014